# Lab Report: Training dynamics of deep models

## 1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

## 2 Results

Figure 1 shows the accuracy by dataset.

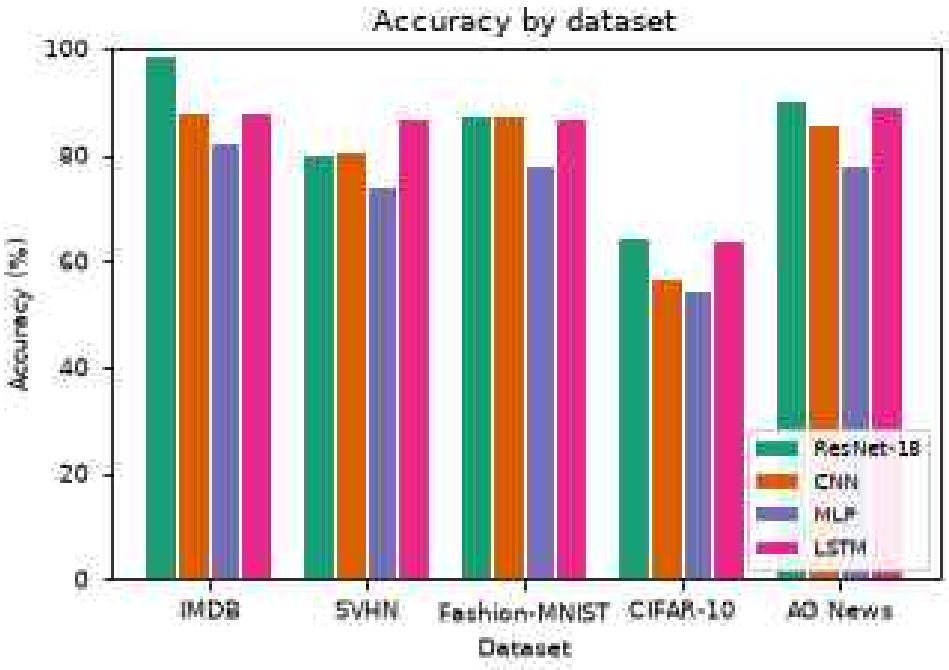



Figure 1: Accuracy by dataset for the models in this experiment.

MLP achieves over 75% accuracy on Fashion-MNIST. On IMDB, ResNet-18 is 11 points better than LSTM.

Figure 2 shows the accuracy during training.

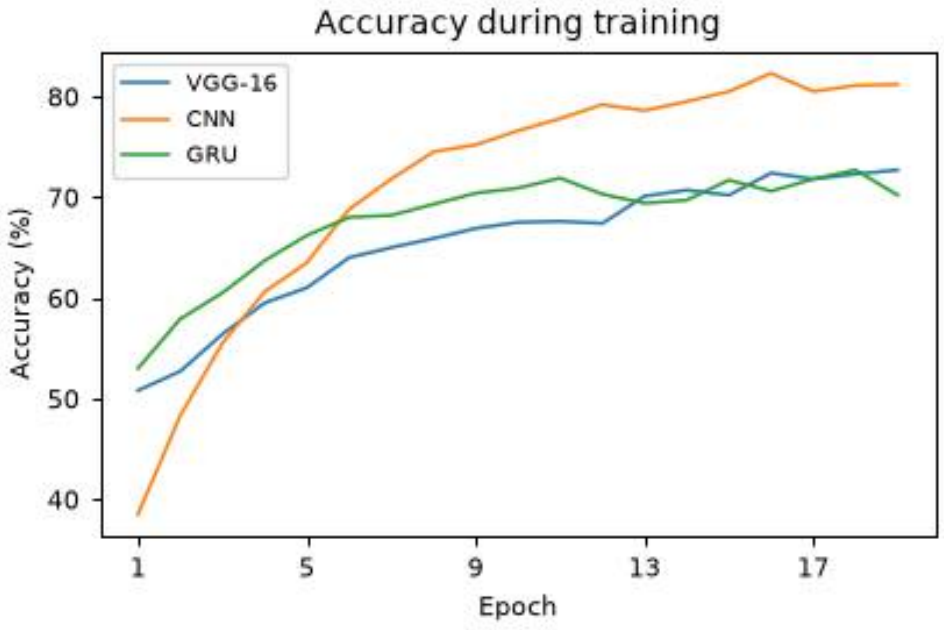



Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 2, GRU achieves about 60% accuracy. CNN outperforms GRU by 5.7

percentage points at epoch 10.

## 3 Discussion

On CIFAR-10, Naive Bayes obtains an accuracy of 54.7%. We note that the training was stable for all models.